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Testing for Sufficient Follow-Up in Censored Survival Data by Using Extremes

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ABSTRACT

In survival analysis, it often happens that some individuals, referred to as cured individuals, never experience the event of interest. When analyzing time-to-event data with a cure fraction, it is crucial to check the assumption of "sufficient follow-up," which means that the right extreme of the censoring time distribution is larger than that of the survival time distribution for the noncured individuals. However, the available methods to test this assumption are limited in the literature. In this article, we study the problem of testing whether follow-up is sufficient for light-tailed distributions and develop a simple novel test. The proposed test statistic compares an estimator of the noncure proportion under sufficient follow-up to one without the assumption of sufficient follow-up. A bootstrap procedure is employed to approximate the critical values of the test. We also carry out extensive simulations to evaluate the finite sample performance of the test and illustrate the practical use with applications to leukemia and breast cancer data sets.

1 | Introduction

In clinical studies, an increasing number of patients diagnosed with a specific cancer will never die of their cancer or will never experience a relapse after receiving treatment. Those individuals are referred to as immune or cured, while the remaining individuals are often called susceptible. In reality, the phenomenon of individuals being cured is quite common especially in cancer clinical studies, mainly thanks to the significant medical advancements that have been made. For example, breast cancer mortality experienced a 39% decline between 1989 and 2015 (DeSantis et al. 2017). Siegel et al. (2023) also showed that the 5-year relative survival rate for breast cancer is 91% for patients diagnosed during 2012–2018. Some patients with breast cancer will never experience a relapse or recurrence of breast cancer after receiving treatment and thus be cured. In Section 4, we consider the case of breast cancer in more detail and are interested in identifying correctly their cure rate. We also analyze data on leukemia patients, whose cure rates have also increased steadily over the last years thanks to advanced technology to treat this cancer. Other examples of situations where some subjects are immune to the event of interest can be found in sociology (the time to become a parent), criminology (the time to commit crimes for a portion of the released population), and economics (the time to find a new job after a layoff), among others.

Cure models are frequently employed to analyze this type of data due to their ability to take the cure fraction into account. The literature on cure models covers a wide range of research areas, including the estimation of the effects of covariates, the cure proportion, and the survival function, and tests for the presence of cured individuals and for the validity of the assumed cure models,

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among others. We refer the reader to Peng and Taylor (2014), Amico and Van Keilegom (2018), Legrand and Bertrand (2019), and Maller et al. (2024) for recent review papers on cure models, and to Maller and Zhou (1996) and Peng and Yu (2021) for books on cure models.

When analyzing survival data with a cure fraction, it is essential to test the assumption of sufficient follow-up, which means that the right extreme of the censoring time distribution is larger than that of the survival time distribution for the susceptibles. In practice, the length of the plateau in the Kaplan–Meier estimator (KME, Kaplan and Meier 1958) of the survival function and the number of censored observations contained in that plateau can serve as useful indicators to test the assumption of sufficient follow-up. However, this judgment tends to be rather subjective and ambiguous. To the best of our knowledge, there is still a scarcity of research on testing for sufficient follow-up in the statistical literature and the few existing papers on this topic all suffer from serious problems, which makes them difficult to use in practice. Based on the interval length from the largest uncensored observed survival time to the largest observed survival time, Maller and Zhou (1992) first proposed a testing procedure to this particular testing problem, and this test was studied in detail by Maller and Zhou (1994). However, their proposed test only estimates the significance level but does not control it, and the numerical study given in Maller and Zhou (1996) showed that their test is too conservative. Modifications for Maller and Zhou (1992)'s test were provided by Maller and Zhou (1996) and Shen (2000), respectively. Klebanov and Yakovlev (2007) discussed the aforementioned three testing procedures for the problem of testing for sufficient follow-up and found out that they fail to produce satisfactory results due to the highly unstable nature of the KME and related statistics in the presence of censoring. Recently, Maller, Resnick, and Shemehsavar (2024) filled in the gap in the asymptotic properties of the test statistic proposed by Maller and Zhou (1996), while the distributions of the tests proposed by Maller and Zhou (1992) and Shen (2000) remain to be investigated. The simulation studies we conduct in Section 3 reveal that the empirical levels of the test proposed by Maller and Zhou (1996) are significantly higher than the nominal level especially when the follow-up period increases, and thus this testing procedure may incorrectly check the problem of testing for sufficient follow-up.

Whether follow-up is sufficient or not has a significant impact on the estimation of the cure proportion. When the survival time is exposed to random right-censoring, the noncured individuals can either fail or be censored, whereas cured individuals will all be censored. Hence, a complicated problem arises regarding the identification of the cure rate due to the lack of data in the right tail of the survival function. A solution to this problem is to impose the assumption of sufficient follow-up. Under this assumption, Maller and Zhou (1992) suggested a consistent nonparametric estimator of the cure rate that corresponds to the height of the plateau of the KME of the survival function. In addition, Xu and Peng (2014) further proposed a nonparametric estimator of the cure rate in the presence of covariates and established its consistency and asymptotic normality.

When the assumption of sufficient follow-up is not satisfied, however, the nonparametric estimators proposed by Maller and Zhou (1992) and Xu and Peng (2014) will overestimate the cure rate, since the data do not provide enough information on the right tail of the survival function. To tackle this problem, Escobar-Bach and Van Keilegom (2019) first considered extrapolation techniques from extreme value theory (de Haan and Ferreira 2006) to avoid the assumption of sufficient follow-up and corrected the estimator proposed by Maller and Zhou (1992), which has also been extended to the case where covariates are present by Escobar-Bach and Van Keilegom (2023). The vital assumption imposed by Escobar-Bach and Van Keilegom (2019) is that the survival function of the noncured individuals belongs to the Fréchet domain of attraction. Escobar-Bach et al. (2022) further considered the case where the survival function of the noncured individuals is in the Gumbel domain of attraction that covers most of the parametric distributions assumed in practice for analyzing survival data, including Weibull, lognormal, and Gamma distributions, among others. Likewise, we assume that the distribution of the noncured individuals belongs to the Gumbel domain of attraction in this article.

Extreme value theory is frequently utilized in the literature on survival analysis with right-censored data. In addition to the aforementioned articles that applied extreme value theory to estimate the cure rate, extreme value theory is also considered to estimate the extreme value index (see, e.g., Beirlant and Guillou 2001; Einmahl, Fils-Villetard, and Guillou 2008; Worms and Worms 2014; Stupfler 2016). Recently, Maller and Resnick (2022) developed the asymptotic distributions of statistics relevant to testing for the presence of cured individuals in the population through the application of extreme value theory. Maller, Resnick, and Shemehsavar (2022) further studied the asymptotic distributions of the largest uncensored and largest observed survival times for survival data with a cure fraction. Here, we will show that it becomes possible to construct a new test for the assumption of sufficient follow-up by using extreme value theory.

The motivation for proposing a novel testing procedure is twofold. First, as already mentioned earlier, all existing tests suffer from serious problems with the type I error. Some of them are very conservative, while others are systematically anticonservative. We hope to overcome this by using a different approach. Second, the null hypothesis of all of the aforementioned tests is the hypothesis of insufficient follow-up. By doing so, these tests control the probability of erroneously concluding that the support is sufficient, which is important if we want to make sure that the cure rate is not overestimated. However, in certain situations we might want to make sure that a study can be safely ended as soon as the cure rate can be correctly identified, since from a cost or economic perspective this might be a primary concern. In that case we should control the probability of erroneously concluding that the support is insufficient, which we will do by defining the null hypothesis as the hypothesis of sufficient follow-up. Instead of being based on the difference between the largest uncensored and the largest observed survival times, the proposed test statistic involves the comparison of two nonparametric estimators of the cure rate. We reject the null hypothesis of sufficient follow-up if the value of the test statistic is large. Moreover, the asymptotic distribution of our test statistic is shown to be a normal distribution under the null hypothesis. Extensive simulation studies confirm that our proposed test performs satisfactorily for finite samples. In particular, under insufficient follow-up, our test still

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works well in comparison to Maller and Zhou (1996)'s testing procedure.

The article is organized as follows. In Section 2, we describe in detail the proposed testing procedure and show some asymptotic results. Besides, a bootstrap procedure is also employed to approximate the critical values of the test in this section. Simulation studies are reported in Section 3 to evaluate the finite-sample performance of the test. In Section 4, we illustrate the practical applicability through the analysis of both leukemia and breast cancer data sets. Section 5 presents a discussion.

2 | Methodology

2.1 | Testing Procedure

We consider a random right censorship model, and let T be the survival time and C be the random right-censoring time. The distribution functions of T and C are denoted as F and F_c, respectively. The population is divided into two groups consisting of the noncured and cured individuals with proportions p and $1-p$, $0<p<1$. The cured individuals, also called immune, never experience the event of interest and hence have by convention an infinite survival time. This leads to an improper distribution F with total mass strictly less than 1 and the representation

$$F(t)=\mathbb{P}(T\le t)=pF_0(t), \tag{1}$$

where F_0 is a proper distribution function for the noncured population, also called susceptibles. The value of p actually represents the probability $\mathbb{P}(T<+\infty)$ while $1-p$ can be seen as the proportion of immunes, called the cure rate. The presence of right-censoring only allows us to observe the couple (Y,δ), where $Y=\min(T,C)$ and $\delta=\mathbb{1}(T\le C)$. Due to some identification issues, we impose the common assumption that T and C are independent. This particularly ensures that the distribution function H of the observed random time Y satisfies $1-H=(1-F)(1-F_c)$. In the sequel, we denote for convenience $\bar{G}=1-G$ and $\tau_G=\sup\{t\ge 0: G(t)<1\}$ as the right extreme for any generic distribution function G. For example, for the distribution function H, $\bar{H}=1-H$, and $\tau_H=\sup\{t\ge 0: H(t)<1\}$. Obviously, $\tau_H=\min(\tau_{F_c},\tau_F)=\tau_{F_c}$ since $\tau_F=\infty$.

Let $\{(Y_i,\delta_i)\}_{1\le i\le n}$ be a sample of n independent and identically distributed (i.i.d.) random pairs drawn from the distribution of (Y,δ), and $\tilde{t}_{(1)}<\tilde{t}_{(2)}<\dots<\tilde{t}_{(K)}$ represent K distinct uncensored survival times. Maller and Zhou (1992) suggested an estimator of p based on (1) by returning the maximum value of an estimate of the distribution F. Specifically, they proposed the following nonparametric estimator:

$$\hat{p}_n=\hat{F}_n(t_{(n)}), \tag{2}$$

where $t_{(n)}=\max\{Y_i, i=1,\dots,n\}$ is the largest observed time in the sample, $\hat{F}_n(t)$ is the KME of F defined as

$$\hat{F}_n(t)=1-\prod_{\tilde{t}_{(i)}\le t}\left(1-\frac{d_i}{D_i}\right),$$

D_i is the number of individuals at risk at $\tilde{t}_{(i)}$, and d_i is the number of individuals having failure at $\tilde{t}_{(i)}$. As discussed in Maller and Zhou (1992), $\hat{p}_n$ is consistent if and only if the assumption of sufficient follow-up is valid, which is equivalent in this case to $\tau_{F_0}\le\tau_{F_c}$. The case of insufficient follow-up however, when $\tau_{F_0}>\tau_{F_c}$, yields an underestimation of p by $\hat{p}_n$.

Recently, Escobar-Bach et al. (2022) proposed a nonparametric estimator $\hat{p}_G(\epsilon)$ of p when $\bar{F}_0$ is in a class of particular slowly varying functions, where the definition of ϵ is given below. More precisely, they assumed that the susceptible distribution is in the Gumbel domain of attraction. This condition invokes the tail behavior of the distribution and considers that there exists a positive function l such that for all $y\in\mathbb{R}$,

$$\lim_{t\to\tau_{F_0}}\frac{\bar{F}_0(t+yl(t))}{\bar{F}_0(t)}=e^{-y}. \tag{3}$$

In their paper, they also imposed a condition, namely, the von Mises condition (Resnick 1987, p. 42, eq. 1.8), for the distribution F_0. This condition is a sufficient but not necessary condition for a distribution belonging to a domain of attraction. For the Gumbel domain of attraction, the von Mises condition says that F_0 has a finite second derivative F_0'' in a neighborhood $(\tau_{F_0}-\epsilon,\tau_{F_0})$ for some $\epsilon\in(0,\tau_{F_0})$ with

$$\lim_{t\to\tau_{F_0}}\frac{F_0''(t)\bar{F}_0(t)}{\{F_0'(t)\}^2}=-1. \tag{4}$$

In practice, most of the distributions in the Gumbel domain satisfy the von Mises condition, such as Weibull, lognormal, and Gamma distributions. The nonparametric estimator is then given by

$$\hat{p}_G(\epsilon)=\hat{F}_n(t_{(n)}-\epsilon)+\frac{\{\hat{F}_n(t_{(n)}-\epsilon/2)-\hat{F}_n(t_{(n)}-\epsilon)\}^2}{2\hat{F}_n(t_{(n)}-\epsilon/2)-\hat{F}_n(t_{(n)}-\epsilon)-\hat{F}_n(t_{(n)})}. \tag{5}$$

If $\tau_{F_c}<\tau_{F_0}$ and the von Mises condition holds, Theorem 2 of Escobar-Bach et al. (2022) shows that the latter estimator $\hat{p}_G(\epsilon)$ converges in probability to

$$p_G(\epsilon)=F(\tau_H-\epsilon)+\frac{\{F(\tau_H-\epsilon/2)-F(\tau_H-\epsilon)\}^2}{2F(\tau_H-\epsilon/2)-F(\tau_H-\epsilon)-F(\tau_H)} \tag{6}$$

as $n\to\infty$, and at the same time, converges to p as $\epsilon\to 0$ and $\tau_{F_c}\to\tau_{F_0}$. Here, we require that $\epsilon<\tau_{F_c}$. Roughly speaking, the idea relies on correcting the estimator $\hat{p}_n$ by some additive statistics based on the extreme value condition, compensating the underestimation of $\hat{p}_n$ under insufficient follow-up. Reversely, this latter correction turns out to be almost null under sufficient follow-up.

Regardless of whether follow-up is sufficient, we always have $\tau_H=\tau_{F_c}$ since $p<1$. Thus, when $\tau_{F_0}\le\tau_{F_c}$, $F(\tau_H)$ is equal to p. For the range of ϵ, we mentioned before that under insufficient follow-up, ϵ should belong to the interval $(0,\tau_{F_c})$. Under sufficient follow-up, it is easy to prove that $\hat{p}_G(\epsilon)$ converges in probability to (6) as $n\to\infty$. In this case, however, the range of ϵ is $(\tau_{F_c}-\tau_{F_0},\tau_{F_c})$ and we require that $\tau_{F_c}<\infty$, since when $\epsilon\in(0,\tau_{F_c}-\tau_{F_0}]$

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or $\tau_{F_c} = \infty$, the denominator of (6) is equal to 0. Therefore, we impose that ϵ belongs to the interval $\Theta = (0 \vee (\tau_{F_c} - \tau_{F_0}), \tau_{F_c})$. In practice, ϵ can be chosen in the interval $(t_{(n)} - \tilde{t}_{(K)}, t_{(n)})$, where $\tilde{t}_{(K)}$ is the largest uncensored observed survival time.

In this article, we are interested in testing the assumption of sufficient follow-up. The hypotheses are

$$H_0 : \tau_{F_0} \le \tau_{F_c} \quad \text{versus} \quad H_1 : \tau_{F_0} > \tau_{F_c}.$$

The above arguments motivate us to propose a test statistic through the difference between $\hat{p}_n$ and $\hat{p}_G(\epsilon)$. Specifically, we define the test statistic as

$$T_n = \hat{p}_G(\epsilon) - \hat{p}_n.$$

Under sufficient follow-up, T_n will be close to zero while under insufficient follow-up, $\hat{p}_G(\epsilon)$ will be strictly larger than $\hat{p}_n$ and so T_n will be strictly positive. For large values of T_n, we hence expect to reject the null hypothesis H_0. To determine the rejection threshold for the null hypothesis, we will establish in the next subsection the asymptotic normality of T_n under H_0. In addition, selecting an appropriate ϵ is crucial for achieving satisfactory performance, which is discussed in detail in Section 3.

Remark 2.1. Unlike $\hat{p}_n$, the estimator $\hat{p}_G(\epsilon)$ offers an accurate estimate of the noncure rate under both sufficient and insufficient follow-up (Escobar-Bach et al. 2022). Also note that the presence or absence of cured individuals is irrelevant in our testing procedure, since our approach is still valid when $p = 1$.

2.2 | Asymptotic Theory

In this subsection, we will establish the asymptotic normality of the test statistic T_n. The asymptotic normality of the KME of F (Maller and Zhou 1996, Theorem 3.14) plays an important role in proving the asymptotic normality of T_n. To do so, we first introduce some notations. Suppose $Z(t)$ is a Gaussian process on $[0, \tau_H]$ with zero mean and variance function

$$v(t) = \int_{[0,t]} \frac{dF(s)}{\bar{F}(s)\{1 - F(s^-)\}\{1 - F_c(s^-)\}}.$$

We also define $D[0, \tau_H]$ as the functional space of càdlàg real functions on $[0, \tau_H]$ associated with Skorokhod's topology. The asymptotic normality of the KME of F is summarized in the following theorem.

Theorem 2.1 Theorem 3.14 from Maller and Zhou (1996). *Suppose that F is continuous at τ_H in case $\tau_H < \infty$. Suppose furthermore that* $\lim_{t\to\tau_H}\{F(\tau_H) - F(t)\}^2 v(t) = 0$ *and*

$$\lim_{t\to\tau_H} \int_{(t,\tau_H)} \frac{I\{0 \le F_c(s^-) < 1\}\bar{F}(s)}{\{1 - F_c(s^-)\}\{1 - F(s^-)\}} dF(s) = 0. \tag{7}$$

Then, $\lim_{t\to\tau_H} \bar{F}(t)Z(t)$ *exists and is finite almost surely, and for each $t \ge 0$,*

$$X_n(t) = \sqrt{n}\frac{\bar{F}(t)}{\bar{F}(t \wedge t_{(n)})}\{\hat{F}_n(t \wedge t_{(n)}) - F(t \wedge t_{(n)})\} \tag{8}$$

converges weakly in $D[0, \tau_H]$ to $X(t) = I(t \in [0, \infty))\bar{F}(t)Z(t) + I(t = \infty)R$ as $n \to \infty$, where

$$R = \begin{cases} \bar{F}(\tau_H)Z(\tau_H) & \text{if} \quad H(\tau_{H^-}) < 1, \\ \lim\limits_{t\to\tau_H} \bar{F}(t)Z(t) & \text{if} \quad H(\tau_{H^-}) = 1. \end{cases} \tag{9}$$

The following theorem summarizes the asymptotic normality of T_n under sufficient follow-up.

Theorem 2.2. *Suppose that Conditions (C1)–(C3) in the Appendix hold and F_0 satisfies (4). Suppose also that $\tau_{F_c} < 2\tau_{F_0}$. Then, for each $\epsilon \in (2(\tau_{F_c} - \tau_{F_0}), \tau_{F_c})$, the test statistic T_n is asymptotically normal, $\sqrt{n}(T_n - b_\epsilon) \to N(0, \sigma_\epsilon^2)$ in distribution as $n \to \infty$, where the asymptotic bias is*

$$b_\epsilon = F(\tau_H - \epsilon) + \frac{\{F(\tau_H - \epsilon/2) - F(\tau_H - \epsilon)\}^2}{2F(\tau_H - \epsilon/2) - F(\tau_H - \epsilon) - F(\tau_H)} - F(\tau_H),$$

the asymptotic variance is

$$\begin{aligned} \sigma_\epsilon^2 &= \sum_{i=1}^{3}\sum_{j=1}^{3} s_i s_j \bar{F}\{\tau_H + \epsilon(i-3)/2\}\bar{F}\{\tau_H + \epsilon(j-3)/2\} v \\ &\quad \times\{\tau_H + \epsilon(i \wedge j - 3)/2\}, \\ s_1 &= \frac{\{F(\tau_H - \epsilon/2) - F(\tau_H)\}^2}{\{2F(\tau_H - \epsilon/2) - F(\tau_H - \epsilon) - F(\tau_H)\}^2}, \\ s_2 &= \frac{2\{F(\tau_H) - F(\tau_H - \epsilon/2)\}\{F(\tau_H - \epsilon) - F(\tau_H - \epsilon/2)\}}{\{2F(\tau_H - \epsilon/2) - F(\tau_H - \epsilon) - F(\tau_H)\}^2}, \end{aligned}$$

and

$$s_3 = \frac{\{F(\tau_H - \epsilon/2) - F(\tau_H - \epsilon)\}^2}{\{2F(\tau_H - \epsilon/2) - F(\tau_H - \epsilon) - F(\tau_H)\}^2} - 1.$$

Condition (C2) says that $\int_{[0,\tau_{F_0}]} dF_0(s)/\{1 - F_c(s^-)\} < \infty$, which implies that $\tau_{F_0} \le \tau_{F_c}$. Theorem 2.2 establishes the asymptotic normality of T_n under sufficient follow-up. Its proof is based on Theorem 2.1 and outlined in the Supplementary material. By the proof of Theorem 2.2, we have

$$\sqrt{n}(T_n - b_\epsilon) = s_1 X_n(t_{(n)} - \epsilon) + s_2 X_n(t_{(n)} - \epsilon/2) + s_3 X_n(t_{(n)}) + o_p(1).$$

Under sufficient follow-up, the range of ϵ is $\Theta = ((\tau_{F_c} - \tau_{F_0}), \tau_{F_c})$. The asymptotic bias b_ϵ converges to 0 under H_0 as ϵ goes to $2(\tau_{F_c} - \tau_{F_0})$. When ϵ falls within the interval $(\tau_{F_c} - \tau_{F_0}, 2(\tau_{F_c} - \tau_{F_0})]$, b_ϵ, s_1, s_2, and s_3 are all equal to zero. In this case, $\sqrt{n}(T_n - b_\epsilon) = \sqrt{n}T_n = o_p(1)$. Therefore, according to Theorem 2.2, the asymptotic distribution of T_n under sufficient follow-up is non-degenerate only when $2(\tau_{F_c} - \tau_{F_0})$ is strictly less than τ_{F_c} (i.e., $\tau_{F_c} < 2\tau_{F_0}$) and $\epsilon \in (2(\tau_{F_c} - \tau_{F_0}), \tau_{F_c})$. This condition for $\tau_{F_c} < 2\tau_{F_0}$ can also be found in Theorem 2 in Maller, Resnick, and Shemehsavar (2024). When $2\tau_{F_0} \le \tau_{F_c}$, the asymptotic distribution of T_n is degenerate with $\sqrt{n}T_n = o_p(1)$. In practice, when $2\tilde{t}_{(K)} < t_{(n)}$, T_n is almost surely equal to zero 0. Therefore, our test remains effective in this case and does not rely on any asymptotic distribution.

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Through the preceding discussion, we successfully established the asymptotic normality of the test statistic. However, for small sample sizes, the normal approximation given in Theorem 2.2 does not perform well. Therefore, a bootstrap procedure is used to approximate the critical values of the test. The proposed bootstrap procedure comprises the following sequential steps:

1. Use the original sample to calculate the test statistic T_n.
2. Draw B naive bootstrap samples with replacement from the original sample.
3. For $b = 1, \dots, B$, calculate the bootstrap test statistic T_n^b by using the bth naive bootstrap sample.
4. The critical value of the test is approximated by the $(1-\alpha)$th quantile of $\{T_n^1 - T_n, \dots, T_n^B - T_n\}$, where α is the level of the test.

The proposed bootstrap procedure aligns with that outlined in Bickel and Ren (2001) and the calculated critical values using it consistently approach 0. Thus, the proposed bootstrap procedure is feasible based on the fact that under H_0, T_n is close to zero while under H_1, T_n is strictly positive. More details about the proposed bootstrap method can be found in Bickel and Ren (2001).

3 | Simulation Studies

In the following simulation studies, we consider three distributions for the failure times of the susceptible patients:

i. A Weibull distribution with shape parameter 1.5 and scale parameter 1.5;
ii. An exponential distribution with parameter 1; and
iii. A standard lognormal distribution.

In addition, we generate the censoring variable from a Weibull distribution with shape parameter 1 and scale parameter $\lambda = 1.5, 2, 2.5, 3$ under the null hypothesis, and from uniform distributions $U[0, \mu]$ with $\mu = 2, 2.5, 3, 3.5$ under the alternative hypothesis. It is readily apparent that smaller values of λ correspond to more censoring. Similarly, smaller values of μ mean a shorter follow-up period while large values represent light-censoring. Two values for the noncure rate have been chosen with p equal to 0.7 or 0.9, representing moderate and high noncure rates. Four sample sizes are considered with $n = 400, 800, 1200, 1800$. The nominal level equals $\alpha = 0.05$ and all simulation results are based on 1000 simulation runs.

To compute T_n and obtain satisfactory performance, it is essential to select a suitable ϵ. To do so, Escobar-Bach et al. (2022) considered a quadratic errors criterion to select the optimal ϵ, which leads to favorable results. However, we found that this criterion appears to be ineffective for our test. We also attempted other methods to determine the optimal ϵ, including a double-bootstrap procedure from Cao and Van Keilegom (2006), and a bootstrap method from Escobar-Bach and Van Keilegom (2019). We finally found that when the value of ϵ is close to $t_{(n)}$, the empirical level and empirical power perform well. We show the detailed simulation results for different ϵ in the Supplementary material. It is worthwhile noting that when $\epsilon \in (t_{(n)} - \tilde{t}_{(K)}, 2(t_{(n)} - \tilde{t}_{(K)})]$, T_n always equals 0. Thus, we select ϵ based on two cases: whether $t_{(n)}$ is greater than $2(t_{(n)} - \tilde{t}_{(K)})$ or not. When $2(t_{(n)} - \tilde{t}_{(K)}) < t_{(n)}$, we take ϵ as $\epsilon^* = \frac{9}{8} t_{(n)} - \frac{1}{4} \tilde{t}_{(K)}$ and otherwise fix it to $t_{(n)}$, where $\epsilon^* = 2(t_{(n)} - \tilde{t}_{(K)}) + \phi(t_{(n)} - 2(t_{(n)} - \tilde{t}_{(K)}))$ and $\phi = 7/8$. When ϕ is assigned other values close to 1, our simulation continues to perform effectively. Besides, if the denominator of $\hat{p}_G(\epsilon^*)$ equals 0 or $\hat{p}_G(\epsilon^*) < \hat{p}_n$, we redefine $\hat{p}_G(\epsilon^*) = \hat{p}_n$ and hence set $T_n = 0$. In addition, $\hat{p}_G(\epsilon^*)$ is redefined as 1 if $\hat{p}_G(\epsilon^*) > 1$.

We evaluate the performance of the test through its empirical level and power, where the bootstrap critical value is obtained from 500 bootstrap iterations. Tables 1–3 summarize the results for three considered distributions. In Table 1 under H_0, for a large λ, the empirical level is quite close to the nominal level, even for the small sample size ($n = 400$) and moderate censoring rate ($p = 0.7$). Besides, for a small λ, the empirical level is slightly higher than the nominal level when the sample size is small and the censoring rate is lower. For example, when $\lambda = 1.5$, $n = 400$, and $p = 0.9$, the empirical level is 0.121 while the empirical level quickly reduces to the nominal level when either the sample size or λ increase. This phenomenon is reasonable since, the smaller value of λ corresponds to the larger censoring rate. For example, when $\lambda = 3$, the censoring rate is 27% at $p = 0.9$. As λ decreases to 1.5, the censoring rate reaches 39%. Furthermore, under H_1 we observe that for each setting, the empirical power increases to 1 as the sample size gets larger. For small sample sizes, however, the empirical power sometimes falls short of expectations. For example, when $\mu = 3.5$, $p = 0.7$, and $n = 400$, the empirical power is 0.125, which is far from 1. A larger sample size leads to a substantial increase in the empirical power. Besides, the empirical power also improves with decreasing μ since smaller μ means larger departures from the null hypothesis.

In Tables 2–3, we observe that the empirical level is similar to that in Table 1. In addition, for a high noncure rate, the empirical powers in Tables 2–3 also quickly tend to 1 as the sample size increases like those in Table 1. However, for a moderate noncure rate, compared to the empirical powers shown in Table 1, the empirical powers in Tables 2–3 approach 1 at a slightly slower rate as the sample size increases. The reason for this is that the censoring rate in Tables 2–3 is larger than that in Table 1 for each respective setting.

As previously mentioned in Section 1, there are several tests in the statistical literature related to this testing problem. However, their testing procedures are not suitable for comparison with our proposed test as they all considered insufficient follow-up as the null hypothesis, but the null hypothesis in this article is sufficient follow-up. Nevertheless, we still show the simulation results of Maller and Zhou (1996)'s testing procedure to evaluate the performance of their test. Unlike our proposed test, their testing procedure is based on the length of the interval $[\tilde{t}_{(K)}, t_{(n)}]$. Specifically, their test statistic is

$$Q_n = \frac{1}{n} \#\{\text{uncensored observations in } [2\tilde{t}_{(K)} - t_{(n)}, \tilde{t}_{(K)})\}.$$

When a large value of Q_n is observed, the null hypothesis of insufficient follow-up is rejected, indicating that follow-up is sufficient. The asymptotic properties of Q_n were discussed in detail in Maller, Resnick, and Shemehsavar (2024). Table 4 summarizes

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TABLE 1 | Simulation results of our test, when the failure time of susceptible patients follows a Weibull distribution with shape parameter 1.5 and scale parameter 1.5.

		$\lambda = 3$		$\lambda = 2.5$		$\lambda = 2$		$\lambda = 1.5$	
		$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$
H_0	n	27% cens	43% cens	30% cens	45% cens	33% cens	48% cens	39% cens	52% cens
	400	0.044	0.059	0.041	0.052	0.047	0.051	0.121	0.049
	800	0.039	0.047	0.045	0.061	0.039	0.047	0.058	0.038
	1200	0.054	0.045	0.035	0.054	0.031	0.035	0.039	0.048
	1800	0.047	0.059	0.045	0.043	0.045	0.051	0.028	0.029
		$\mu = 3.5$		$\mu = 3$		$\mu = 2.5$		$\mu = 2$	
		$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$
H_1	n	28% cens	44% cens	31% cens	46% cens	34% cens	49% cens	40% cens	53% cens
	400	0.717	0.125	0.858	0.397	0.927	0.632	0.849	0.657
	800	0.924	0.403	0.974	0.487	0.979	0.738	0.937	0.801
	1200	0.991	0.816	0.992	0.827	0.988	0.865	0.970	0.878
	1800	0.999	0.935	0.994	0.969	0.985	0.985	0.981	0.956

Note: H_0, the null hypothesis; H_1, the alternative hypothesis; n, the sample size; p, the noncure rate; λ, the parameter of the censoring distribution under H_0; μ, the parameter of the censoring distribution under H_1.

TABLE 2 | Simulation results of our test, when the failure time of susceptible patients follows an exponential distribution with parameter 1.

		$\lambda = 3$		$\lambda = 2.5$		$\lambda = 2$		$\lambda = 1.5$	
		$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$
H_0	n	33% cens	47% cens	36% cens	50% cens	40% cens	53% cens	46% cens	58% cens
	400	0.043	0.044	0.052	0.043	0.058	0.046	0.086	0.036
	800	0.053	0.048	0.048	0.045	0.036	0.048	0.050	0.031
	1200	0.041	0.040	0.045	0.058	0.030	0.050	0.032	0.042
	1800	0.053	0.048	0.051	0.051	0.032	0.037	0.026	0.034
		$\mu = 3.5$		$\mu = 3$		$\mu = 2.5$		$\mu = 2$	
		$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$
H_1	n	35% cens	50% cens	38% cens	52% cens	43% cens	56% cens	49% cens	60% cens
	400	0.214	0.025	0.333	0.045	0.430	0.104	0.553	0.204
	800	0.514	0.052	0.628	0.108	0.709	0.116	0.753	0.183
	1200	0.821	0.431	0.906	0.489	0.934	0.462	0.930	0.407
	1800	0.960	0.757	0.986	0.789	0.990	0.762	0.991	0.748

Note: H_0, the null hypothesis; H_1, the alternative hypothesis; n, the sample size; p, the noncure rate; λ, the parameter of the censoring distribution under H_0; μ, the parameter of the censoring distribution under H_1.

the simulation results of Q_n when the distribution of the failure time for susceptibles is a Weibull distribution. To avoid ambiguity, in Table 4, we denote $\tilde{H}_0$ and $\tilde{H}_1$ as the null hypothesis of insufficient follow-up and the alternative hypothesis of sufficient follow-up, respectively. Here, the empirical level and empirical power of their test are obtained based on the proposed bootstrap procedure in this article. The results in Table 4 indicate an unsatisfactory performance of their test since all empirical levels are significantly greater than the nominal level, even in the case of large sample sizes. Besides, these empirical levels do not appear to decrease as the sample size increases, which seems to suggest that the performance of Q_n will not be enhanced by increasing the sample size.

4 | Real Data Analysis

4.1 | Leukemia Data Analysis

Leukemia is the most prevalent form of cancer among children and ranks as the seventh primary cause of cancer-related mortality in the United States. At present, the survival rates

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TABLE 3 | Simulation results of our test, when the failure time of susceptible patients follows a standard lognormal distribution.

		$\lambda = 3$		$\lambda = 2.5$		$\lambda = 2$		$\lambda = 1.5$	
		$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$
H_0	n	41% cens	54% cens	44% cens	57% cens	50% cens	61% cens	56% cens	66% cens
	400	0.056	0.034	0.071	0.043	0.096	0.046	0.157	0.049
	800	0.035	0.035	0.062	0.032	0.072	0.024	0.111	0.042
	1200	0.036	0.034	0.037	0.025	0.051	0.030	0.092	0.032
	1800	0.037	0.031	0.038	0.029	0.053	0.019	0.088	0.027
		$\mu = 3.5$		$\mu = 3$		$\mu = 2.5$		$\mu = 2$	
		$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$
H_1	n	45% cens	57% cens	49% cens	60% cens	54% cens	64% cens	60% cens	69% cens
	400	0.264	0.044	0.305	0.074	0.399	0.116	0.482	0.196
	800	0.576	0.110	0.606	0.114	0.620	0.147	0.642	0.170
	1200	0.851	0.495	0.885	0.510	0.898	0.452	0.879	0.342
	1800	0.956	0.783	0.970	0.797	0.972	0.753	0.979	0.705

Note: H_0, the null hypothesis; H_1, the alternative hypothesis; n, the sample size; p, the noncure rate; λ, the parameter of the censoring distribution under H_0; μ, the parameter of the censoring distribution under H_1.

TABLE 4 | Simulation results of the test proposed by Maller and Zhou (1996), when the failure time of susceptible patients follows a Weibull distribution with shape parameter 1.5 and scale parameter 1.5.

		$\mu = 3.5$		$\mu = 3$		$\mu = 2.5$		$\mu = 2$	
		$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$
$\tilde{H}_0$	n	28% cens	44% cens	31% cens	46% cens	34% cens	49% cens	40% cens	53% cens
	400	0.166	0.171	0.154	0.167	0.158	0.180	0.183	0.150
	800	0.146	0.141	0.161	0.168	0.189	0.168	0.187	0.160
	1200	0.179	0.140	0.177	0.163	0.183	0.181	0.194	0.155
	1800	0.170	0.149	0.178	0.176	0.185	0.173	0.189	0.201
		$\lambda = 3$		$\lambda = 2.5$		$\lambda = 2$		$\lambda = 1.5$	
		$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$	$p = 0.9$	$p = 0.7$
$\tilde{H}_1$	n	27% cens	43% cens	30% cens	45% cens	33% cens	48% cens	39% cens	52% cens
	400	0.928	0.997	0.843	0.986	0.697	0.945	0.510	0.828
	800	0.979	0.999	0.934	0.993	0.823	0.982	0.590	0.887
	1200	0.995	1.000	0.969	0.999	0.864	0.990	0.624	0.897
	1800	0.998	1.000	0.983	1.000	0.916	0.993	0.655	0.909

Note: $\tilde{H}_0$, the null hypothesis; $\tilde{H}_1$, the alternative hypothesis; n, the sample size; p, the noncure rate; μ, the parameter of the censoring distribution under $\tilde{H}_0$; λ, the parameter of the censoring distribution under $\tilde{H}_1$.

of leukemia patients are steadily increasing. Siegel et al. (2023) showed the 5-year survival rates of leukemia patients for three distinct time periods: 1975–1977, 1995–1997, and 2012–2018, which were determined to be 34%, 48%, and 66%, respectively. The significant improvement in leukemia patients' 5-year survival rates can be attributed to the advancements in medical standards. For example, the chimeric antigen receptor (CAR) T cells technology (June et al. 2018), the most advanced technology available to treat leukemia, has made rapid advancements and received commercial approvals recently. In the future, we can expect that more patients will never experience a relapse or recurrence of leukemia after treatment.

Here, we will analyze the leukemia data from the Surveillance, Epidemiology, and End Results (SEER) database to illustrate the proposed testing procedure. The SEER database collects cancer case data from population-based cancer registries that recorded approximately 47.9% of the US population, and has been utilized by numerous researchers, statisticians, and policy makers in the United States. Besides, an annual update keeps the SEER

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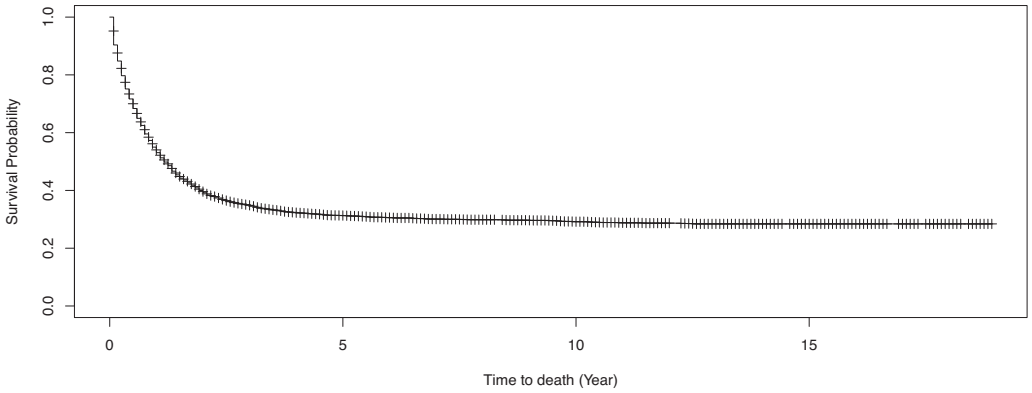

FIGURE 1 | The Kaplan–Meier curve for the leukemia data.

database current. The complete SEER database can be obtained from the website https://seer.cancer.gov/. The information the SEER registers collected includes patient demographics such as gender and age, year of diagnosis, primary tumor site, stage at diagnosis, follow-up time, and vital status, among others. The leukemia data we will analyze are extracted from the database "Incidence-SEER Research Data, 18 Registries, Nov 2020 Sub (2000–2018)."

The follow-up time is the duration from the diagnosis of leukemia to the death caused by leukemia; times to death due to other causes than leukemia or the end of follow-up are considered as censored. We only consider the patients with acute monocytic leukemia, and exclude patients with zero or unknown survival time and those with unknown vital status recode. After excluding the patients mentioned above, the cohort of interest consists of 2624 patients out of which 1416 are male and 1208 are female. The censoring rate is 35.4%, and the uncensored survival time ranges from 0.08 to 12.50 years, with a median of 0.58 years.

We first present the Kaplan–Meier curve in Figure 1, where the symbol "+" on the curves indicates censored individuals. The Kaplan–Meier curve levels off at about 0.283 and has a long plateau. Besides, a total of 210 patients have censored survival times exceeding the largest uncensored survival time. Hence, it appears that the presence of a cure fraction is plausible and the assumption of sufficient follow-up holds. Next, we present an illustrative analysis of the leukemia data to demonstrate the proposed testing procedure. As in the simulations, we set ϵ to $\epsilon^* = \frac{9}{8}t_{(n)} - \frac{1}{4}\tilde{t}_{(K)}$ when $2(t_{(n)} - \tilde{t}_{(K)}) < t_{(n)}$; otherwise, we set it to $t_{(n)}$. For the leukemia data, the largest observed time $t_{(n)}$ and the largest uncensored observed time $\tilde{t}_{(K)}$ are 18.92 and 12.50, respectively, and thus $2(t_{(n)} - \tilde{t}_{(K)}) < t_{(n)}$ and $\epsilon = 18.16$. The nominal level α is 0.05 and the bootstrap critical value is obtained from 1000 bootstrap iterations. Calculating $\hat{p}_n$ and $\hat{p}_G(\epsilon^*)$ as 0.7171 and 0.7173, respectively, our test statistic equals 0.0002, which is very close to 0. We further calculate the p-value to be 0.1420. Therefore, we fail to reject the null hypothesis. Finally, as a comparison we also implement the test of Maller and Zhou (1996). It is valuable to highlight that the null hypothesis in Maller and Zhou (1996) is insufficient follow-up. All other settings remain consistent with those in the simulations. We calculate the test statistic Q_n as 0.0095. Besides, the resulting p-value is 0.0006, which is very close to 0. Thus, their results reject the hypothesis of insufficient follow-up, which is consistent with our conclusions.

4.2 | Breast Cancer Data Analysis

Female breast cancer, being one of the most prevalent types of cancer, has gained significant attention from statisticians, doctors, and other researchers (Hunt, Whitman, and Hurlbert 2014; Zhang and Kim 2021). From 1989 to 2015, there was a notable reduction in breast cancer mortality by 39% (DeSantis et al. 2017), thanks to significant advances in medical care. For slowly proliferating tumors, a sufficient recording time is needed to estimate the survival rates. Tai et al. (2005) analyzed the breast cancer data from the SEER database by assuming that the survival time of patients with breast cancer follows a lognormal distribution and determined the minimum required follow-up time for obtaining statistical cure rate estimates. They showed that the minimum required follow-up time is 26.3 years for stage II of breast cancer whereas the actual records of the breast cancer data in the SEER database "Incidence-SEER Research Data, 18 Registries, Nov 2020 Sub (2000-2018)" are restricted, lasting only 19 years. Here, we will apply the breast cancer data from the aforementioned SEER database to illustrate the proposed testing procedure.

We are interested in the time from the diagnosis of breast cancer to the death caused by breast cancer. The censoring indicator is set to 1 if the patient dies from breast cancer, or 0 if the patient is censored. Here, we only consider the black females with breast cancer with tumor grade II and stage II. Patients will be excluded if they have zero or unknown survival time. Besides, we also exclude the patients with unknown cancer stage or unknown vital status recode. After exclusions, there are 8618 breast cancer patients in the cohort of interest. The uncensored survival time ranges from 0.08 to 18.25 years, with a median of 4.33 years, and the censoring rate is 81.3%.

The Kaplan–Meier curve for the breast cancer data is presented in Figure 2. We observe that the Kaplan–Meier curve has a short plateau. An accurate assessment of the problem of testing for sufficient follow-up cannot be made through intuition alone. Thus, we will apply the proposed testing procedure to check whether follow-up is sufficient for this breast cancer data. We

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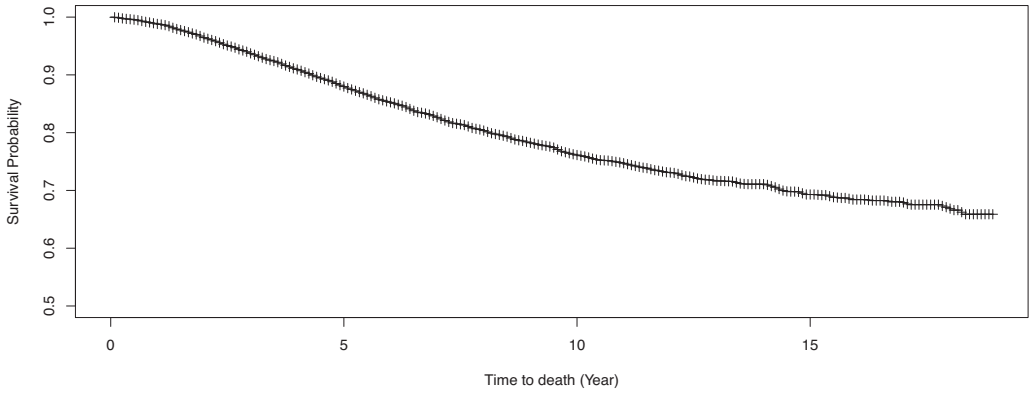

FIGURE 2 | The Kaplan–Meier curve for the breast cancer data.

approximate the critical values in the test by employing the bootstrap procedure shown in Section 2.2 with 1000 bootstrap iterations. Here, the considered nominal level is $\alpha = 0.05$. As in the simulations, when $2(t_{(n)} - \tilde{t}_{(K)}) < t_{(n)}$, we take ϵ as $\epsilon^* = \frac{9}{8}t_{(n)} - \frac{1}{4}\tilde{t}_{(K)}$; otherwise, we fix it to $t_{(n)}$. For the breast cancer data, ϵ is set to 16.72 since $t_{(n)} = 18.92$, $\tilde{t}_{(K)} = 18.25$, and $2(t_{(n)} - \tilde{t}_{(K)}) < t_{(n)}$. The estimators $\hat{p}_n$ and $\hat{p}_G(\epsilon^*)$ are calculated as 0.3437 and 0.4231, which shows that the discrepancy between the naive estimator $\hat{p}_n$ and the estimator $\hat{p}_G(\epsilon^*)$ is quite large. The value of our test statistic is therefore 0.0794. We further calculate the p-value to be 0.0310. From our results, we reject the hypothesis of sufficient follow-up. As a comparison, we further perform the test of Maller and Zhou (1996). The remaining settings align with those used in the simulation. The value of the test statistic Q_n is 0.0002 and Q_n provides a p-value equal to 0.4219. Hence, the hypothesis of insufficient follow-up is not rejected from their results. The results of both testing procedures suggest that the assumption of sufficient follow-up is not met. Thus, to avoid overestimating the cure rate of breast cancer, it is necessary to extend the follow-up period for this breast cancer study.

5 | Discussion

In this article, we proposed a simple novel test to check the assumption of sufficient follow-up. Escobar-Bach et al. (2022) showed that under insufficient follow-up, $\hat{p}_G(\epsilon)$ provides a better estimate of the noncure rate than $\hat{p}_n$. The proposed test statistic is based on the difference between $\hat{p}_G(\epsilon)$ and $\hat{p}_n$. We established the asymptotic normality of our test statistic under the null hypothesis. Extensive numerical studies have shown a satisfactory finite sample performance of our test. All numerical work in this article was implemented in R software, and the R code is available at the website https://github.com/PingXieStat/TestCureSufficientF.

The proposed test is based on the condition that the distribution function F_0 for the noncured population is in the Gumbel domain of attraction (we will simplify by referring to this condition as the "Gumbel condition"). However, there are currently no tests for identifying the Gumbel condition in censored data. Hence, this problem demands additional exploration. We advocate estimating the extreme value index and rejecting the Gumbel condition if the estimate exceeds a given threshold. Besides, when the Gumbel condition is not satisfied, we perform a simulation and the simulation results shown in the Supplementary material suggest that our proposed test still has satisfactory performance when the tail of F_0 is not too heavy.

It is worth pointing out that clinical knowledge about the maximum survival time is important to determine the length of follow-up for an experiment. However, if we want to make sure that a trial can be safely stopped in the sense that the cure rate can be estimated correctly, it is important to be able to assess the condition of sufficient follow-up, since estimates of the cure rate differ depending on whether the sufficient follow-up condition is met or not.

We now point out some interesting future directions. Our proposed testing procedure can be extended to the case when distributions are in the Frechét domain of attraction or the Weibull domain of attraction. The extension of our test is straightforward using our proposed approach by replacing the estimator $\hat{p}_G(\epsilon)$ in T_n with the nonparametric estimator of the noncure rate proposed by Escobar-Bach and Van Keilegom (2019). Besides, in our proposed testing procedure, we only considered the case without covariates. In practice, however, covariates may be included in the model. Hence, the objective of the second research is to extend the proposed testing procedure to the case involving covariates. When covariates are incorporated into the model, Xu and Peng (2014) proposed a nonparametric estimator of the cure rate under sufficient follow-up, while Escobar-Bach and Van Keilegom (2023) further suggested another nonparametric estimator of the cure rate under insufficient follow-up. Thus, the main idea is based on the difference between two nonparametric estimators of the cure rate proposed by Xu and Peng (2014) and Escobar-Bach and Van Keilegom (2023), which gives us an indicator to check the assumption of sufficient follow-up in the presence of covariates. Finally, the choice of epsilon is an important aspect of our testing procedure, and will be studied in more detail in the future.

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helpful discussions and the Surveillance, Epidemiology, and End Results (SEER) Program of the National Cancer Institute for their commendable efforts in the creation of the SEER database. The data sets that support the findings in this paper can be accessed from the website https://seer.cancer.gov/data/. These data sets are used under license for this study and subject to restrictions on availability.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the Surveillance, Epidemiology, and End Results Program. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from https://seer.cancer.gov/data/ with the permission of the Surveillance, Epidemiology, and End Results Program.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

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Appendix

We state the following necessary conditions for Theorem 2.2:

C1. F is continuous at τ_{F_0} in case $\tau_{F_0} < \infty$.

C2. $\int_{[0,\tau_{F_0}]} dF_0(s)/\{1 - F_c(s^-)\} < \infty$.

C3. $\lim_{n\to\infty} n\bar{F}_c(\tau_{F_c} - \xi/\sqrt{n}) = \infty$ for each $\xi > 0$.

In practice, Condition C1 is valid for a wide class of distributions and Condition C2 is a sufficient condition for $\tau_{F_0} \le \tau_{F_c}$. The limit $\lim_{n\to\infty} n\bar{F}_c(\tau_{F_c} - \xi/\sqrt{n}) = \infty$ in Condition C3 holds if the censoring distribution F_c possesses a finite right extreme value τ_{F_c} and has a positive derivative in a left neighborhood of τ_{F_c}. This condition is easily achievable in real-world situations. For instance, the uniform distribution on $[0, \tau]$ has a positive derivative of $1/\tau$ for any $\tau < \infty$ and thus satisfies this condition.

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